

An Example of the Extended Euclidean Algorithm

The (standard) *Euclidean Algorithm* (see, e.g., the excerpt from Bach and Shallit's *Algorithmic Number Theory*) is simply the application of a sequence of integer long divisions. Here's an example of what it might look like for the integers $u_0 := 347$ and $u_1 := 19$:

$$\begin{aligned} 347 &= 18 \cdot 19 + 5 \\ 19 &= 3 \cdot 5 + 4 \\ 5 &= 1 \cdot 4 + 1 \\ 4 &= 4 \cdot 1 + 0 \end{aligned}$$

So, in the notation of Page 68 of our excerpt of Chapter 4 of Bach & Shallit's book, the last equation can be thought of as $u_3 = a_3 u_4$. More simply, there were exactly 4 integer division steps and $\gcd(347, 19) = 1$.

The **Extended** Euclidean Algorithm performs essentially the same calculation, but in a slightly fancier way with matrices. In the end, the computation is no more difficult than what one does for Gaussian Elimination in basic linear algebra. In particular, the divisions steps above suggest the following sequence of elementary integer column operations on the 3×2 matrix on the left below:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 347 & 19 \end{bmatrix} \xrightarrow{-18} \begin{bmatrix} 1 & 0 \\ -18 & 1 \\ 5 & 19 \end{bmatrix} \xrightarrow{-3} \begin{bmatrix} 1 & -3 \\ -18 & 55 \\ 5 & 4 \end{bmatrix} \xrightarrow{-1} \begin{bmatrix} 4 & -3 \\ -73 & 55 \\ \boxed{1} & 4 \end{bmatrix}$$

So we can read off $a=4$ and $b=-73$, simply by looking at the column above the $\boxed{1}$ on the bottom row (of the right-most matrix above). So $4 \cdot 347 + (-73) \cdot 19 = 1$. Note that the multiples we subtract from each column are exactly the negatives of the quotients in our sequence of long divisions above.

Remark #1: ANY sequence of elementary integer column operations that yields a 1 (or $a-1$) on the bottom row would also work. For instance, we could have also done

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 347 & 19 \end{bmatrix} \xrightarrow{-18} \begin{bmatrix} 1 & 0 \\ -18 & 1 \\ 5 & 19 \end{bmatrix} \xrightarrow{-4} \begin{bmatrix} 1 & -4 \\ -18 & 73 \\ 5 & \boxed{-1} \end{bmatrix}$$

which is shorter by one long division. So we obtain $(-4)347 + 73 \cdot 19 = -1$ and, by flipping signs, we get the same a and b as before. $\diamond$

Remark #2: *There are in fact infinitely many pairs $(a, b) \in \mathbb{Z}^2$ with $au + bv = \gcd(u, v)$: For instance, one can easily check for our last example that*

$$4 \cdot 347 + (-73) \cdot 19 = (-15) \cdot 347 + 274 \cdot 19 = 1,$$

and in fact $(5 - 19k)347 + (-55 + 347k)19 = 1$ for any integer k . This is because the solutions $(a, b) \in \mathbb{Z}^2$ of $347a + 19b = 1$ form a (translated) sub-lattice of $\mathbb{Z}^2$ of rank 1. In other words, the solutions are classified by a single integer parameter, starting from any fixed solution. It is also worth noting that there is exactly one solution (a, b) of smallest norm for our example: The solution $(4, -73)$. $\diamond$

One can naturally consider integer linear equations in an arbitrary number of variables, and even arbitrary (finite) collections of integer linear equations. This leads to the concepts of *Hermite Factorization* and *Smith Factorization*. The theory turns out to parallel classical linear algebra very closely and has deep connections to complexity theory and cryptography. For example, *Lattice-based Cryptography* is a highly active area at the intersection of number theory and theoretical computer science.